Enhancing Tropical Cyclone Intensity Estimation from Satellite Imagery through Deep Learning Techniques

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ABSTRACT

This study first utilizes four well-performing pre-trained convolutional neural networks (CNNs) to gauge the intensity of tropical cyclones (TCs) using geostationary satellite infrared (IR) imagery. The models are trained and tested on TC cases spanning from 2004 to 2022 over the western North Pacific Ocean. To enhance the models performance, various techniques are employed, including fine-tuning the original CNN models, introducing rotation augmentation to the initial dataset, temporal enhancement via sequential imagery, integrating auxiliary physical information, and adjusting hyperparameters. An optimized CNN model, i.e., visual geometry group network (VGGNet), for TC intensity estimation is ultimately obtained. When applied to the test data, the model achieves a relatively low mean absolute error (MAE) of 4.05 m s^{-1}. To improve the interpretability of the model, the SmoothGrad combined with the Integrated Gradients approach is employed. The analyses reveal that the VGGNet model places significant emphasis on the distinct inner core region of a TC when estimating its intensity. Additionally, it partly takes into account the configuration of cloud systems as input features for the model, aligning well with meteorological principles. The several improvements made to this model's performance offer valuable insights for enhancing TC intensity forecasts through deep learning.

Key words: tropical cyclone, intensity, deep learning, satellite imagery

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1. Introduction

Tropical cyclones (TCs) stand among the most destructive natural disasters, posing significant threats to human life and property. Accurately estimating the intensity of TCs plays a pivotal role in operational TC forecasting as well as in disaster prevention and mitigation efforts. Given that the intricate physical and dynamic processes related to TC development are not yet fully comprehended (Judt and Chen, 2016), forecasting the intensity of TCs remains a formidable challenge.

Various satellite remote sensing techniques offer practical methods for TC monitoring (Velden et al., 2006; Piñeros et al., 2011; Jin et al., 2014, 2017, 2019; Zhuge et al., 2015; Xu et al., 2016; Zheng et al., 2016). At present, the estimation of TC intensity primarily hinges on the Dvorak technique (Velden et al., 2017), which is based on cloud features in infrared (IR) and visible satellite images. While the Dvorak technique is extensively employed, it is susceptible to errors induced by human judgment and possesses inherent limitations. To bolster the objectivity and automation of IR-based TC intensity analysis, other algorithms have been introduced (Kossin et al., 2007; Olander and Velden, 2007, 2019; Ritchie et al., 2012; Fetanat et al., 2013), such as the digital Dvorak method, advanced Dvorak technique (ADT), objective Dvorak technique (ODT), and advanced ODT (AODT). These computer-driven methodologies diminish the uncertainty and variability in TC intensity estimation and are adopted across various operational TC forecasting centers. However, current techniques for predicting TC genesis and intensity remain notably constrained, largely

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due to reasons such as the less definitive nature of TC genesis and intensity compared to its location, and the intricate physical processes, rendering them challenging to depict accurately through statistical models or dynamic equations.

In recent years, with the extensive application of artificial intelligence, there has been growing attention towards how to apply deep learning (DL) techniques to TC intensity research due to its advantages in image recognition (Li et al., 2020; Zhuo and Tan, 2021). A multilayered deep convolutional neural network (CNN) is initially utilized for classifying TCs by intensity. Using only remote sensing cloud images, the model achieves a superior result with a lower root-mean-square error (RMSE) value (Pradhan et al., 2018). TC intensity is then estimated by using long short-term memory (LSTM) and deep neural networks (DNN), enhancing the strength estimation for catastrophic weather events (Zahera et al., 2019). A multi-model fusion method based on CNN regression is introduced, showcasing the potential of DL in meteorological research (Chen et al., 2018). The ability of DL to discern patterns from data makes it especially suitable for addressing problems with unclear physical mechanisms, such as TC changes. Indeed, the explosive growth of satellite, observational, and reanalysis data offers ample opportunities for DL in TC forecasting.

However, leveraging satellite imagery and DL methods to estimate TC intensity presents distinct challenges. (1) As a type of extreme weather, TCs still present observational challenges that result in limited datasets despite the explosive growth of data in recent years. Due to the limitations imposed by the constrained datasets, it is advisable to utilize transfer learning in combination with fine-tuning methodology (see Section 2.4 for details). This approach contributes to a more effective training process, avoiding the need to initiate a new one using these constrained datasets (Ramdan et al., 2020). (2) Furthermore, TC data significantly deviate from conventional image recognition (Pradhan et al., 2018). Due to temporal and spatial shifts, both the configuration and structural dynamics of a TC undergo transformations. This ever-changing information is not readily discernible from a single IR image, adding complexity to intensity estimations based on satellite imagery (Chen et al., 2020; Wang et al., 2022). Utilizing temporal sequential images could potentially offer enriched contextual insights, amplifying the capacity to grasp the intricate spatial and temporal convection patterns. (3) Additionally, the DL techniques resemble standard image recognition algorithms from computer vision. To provide DL models with meteorological insights that might be missing or difficult to obtain from constrained training datasets, and to reinforce the physical plausibility of models trained via DL, the integration of auxiliary physical knowledge becomes crucial. In light of the potential constraints that these challenges impose on the effectiveness of DL models, this paper is dedicated to addressing these issues to ensure optimized precision in TC intensity prediction.

In this investigation, we design a CNN model to estimate TC intensity over the western North Pacific using satellite imageries based on the above-mentioned improvements. The remainder of the paper is organized as follows. Section 2 introduces the dataset details and preprocessing methodologies. Section 3 illustrates the architecture of the CNN model, its subsequent enhancements, and insights into its interpretability. Finally, our conclusions are consolidated in Section 4.

2. Data and methods

2.1 *Geostationary satellite imagery of TCs*

The observational data obtained from satellite imagery is sourced from the earth's geostationary meteorological Himawari series of satellites. Launched in 1977 by the Japan Meteorological Agency (JMA), the Himawari series represents the first geostationary meteorological satellites. Specifically, the data for 2004 are collected from *Geostationary Operational Environmental Satellite 9* (*GOES-9*), and for 2005–2015 are derived from two Multi-functional Transport SATellites (MTSATs), MTSAT-1R and MTSAT-2. Beginning in 2016, the data source switches to the *Himawari-8* satellite. This new generation satellite in the Himawari series is launched in October 2014 and becomes operational in July 2015 (Bessho et al., 2016; Takeuchi, 2018).

These aforementioned satellites primarily provide IR cloud imagery data. A key advantage of this form of data is its round-the-clock availability, regardless of day or night. This continual data stream facilitates a comprehensive analysis of the entire lifecycle of TCs. Visually, the structure of TCs, accurately captured by these datasets, is highly correlated with the intensity of the convective systems within the TCs. Consequently, utilizing a deep learning-based model for computer vision tasks to assist in convection intensity estimation appears both practical and beneficial.

2.2 *TC data*

Compared to other best track records, the Joint Typhoon Warning Center (JTWC) dataset shows a higher consistency with numerical results and effectively replicates the trends of TC intensity enhancement (Webster et al., 2005; Wu and Zhao, 2012). Thus, this paper uses the best track dataset from 2004 to 2022 over the western North Pacific basin obtained from the JTWC. The data are available each 6-h period, including detailed TC temporal and geographical information, such as time, longitude, latitude, minimum sea level pressure, and maximum wind speed.

The analysis solely considers named TCs with lifetime peak intensities reaching at least 17.85 m s^{-1}. Moreover, extratropical cyclones and TCs after landfall are excluded from the study. To align with the time span of the previously mentioned IR cloud images, a collection of 3335 TC records is extracted.

2.3 *Atmospheric and oceanic data sources*

The fifth-generation ECMWF Re-Analysis (ERA5) data are utilized in this study. The data are employed to compute the vertical wind shear (VWS) and relative humidity (RH) for the western North Pacific. The dataset has a spatial resolution of 0.25° × 0.25° and includes 37 vertical layers spanning from 1000 to 1 hPa. It features a temporal resolution of 6 h. For ocean heat content (OHC), we access the Global Ocean Reanalysis dataset with a 1/12° horizontal resolution (GLORYS12V1) distributed across 50 vertical layers. This dataset furnishes a comprehensive three-dimensional portrayal of oceanic parameters. Produced by Mercator Ocean, it is available via the Copernicus Marine Environment Monitoring Service (CMEMS). Both the atmospheric and oceanic datasets extend consistently from January 2004 to December 2022. The details for calculating these atmospheric and oceanic variables are listed in Table 1, and the justification is extensively discussed in Kaplan et al. (2010). The calculation of OHC is detailed in the Appendix.

2.4 *CNN configuration*

In the 1960s, Hubel and Wiesel (1962) discovered unique neuronal connectivity patterns in a cat's visual cortex, leading to the conception of CNN. Since their victory in the 2012 ImageNet Large Scale Visual Recognition Challenge (ILSVRC; Krizhevsky et al., 2017) have established CNN as the principal algorithm in image processing. Due to its remarkable feature handling and analytic capabilities, CNN has been successfully applied to numerous atmospheric science issues, including the detection of various weather phenomena in observations and climate models (Liu et al., 2016; Lagerquist et al., 2019; McGovern et al., 2019), the representation of subgrid-scale processes (Bolton and Zanna, 2019; Han et al., 2020), and the enhancement of weather predictions and climate projections (Ham et al., 2019; Sønderby et al., 2020; Weyn et al., 2020). In light of the efficiencies and benefits offered by CNNs, we elect to utilize them for indicating the TC intensity from IR satellite cloud imagery.

A CNN comprises three key layers: convolutional, pooling, and dense layers. Convolutional layers contain multiple filters, which enable the extraction of various features from the original input pattern. Pooling layers, typically inserted between successive convolutional layers, help preserve key features while concurrently decreasing the dimensionality of the data. They serve to create invariance to minor shifts and distortions, and increase model generalization while reducing over-fitting (LeCun et al., 2015). Strategically designed convolutional and pooling layers are capable of extracting well-structured features from the raw input data. Dense layers integrate the outcomes of convolution and pooling. CNN is highly suitable for tasks involving image processing and pattern recognition, particularly when dealing with images that possess translational, rotational, and scale invariance. Given that TC images are produced under 5 rotational angles that are centered on the storm center, and typically present well-organized TC structures with different shapes and sizes, it is feasible to assume the aforementioned three invariances.

In meteorological forecasting, it is imperative to note that one cannot blindly apply a specific model, inclusive of its architecture and hyperparameters, to varied application scenarios. Experimentation is pivotal to ascertain the best fit for specific contexts. Given the limitation of data, which is not extensive enough to train a sophisticated model from scratch, a viable solution is to leverage models pre-trained on larger datasets, and subsequently fine-tune them to cater to our tasks (Pan and Yang, 2010). This transfer learning combined with fine-tuning methodology not only abbreviates the training period but, as some studies suggest, can be more effective than starting a new training process (Ramdan et al., 2020).

In our approach to fine-tuning, the models undergo updates during training. These pre-trained models serve as the foundation, negating the need for initializing with random weights. Fine-tuning can be executed across all layers or merely the upper ones. The latter is more favored (Chollet, 2018) since the primary layers in a convolutional network encode versatile, reusable features,

Table 1. Details for calculating the variables as the auxiliary physical information

Predictor	Definition
RH (%)	850–700-hPa relative humidity within a 200–800-km radius
VWS (m s^{-1})	850–200-hPa vertical wind shear within a 500-km radius after vortex removal
OHC (J)	Ocean heat content

whereas the terminal layers capture task-specific features. Adjusting these task-specific features proves more efficient. Conversely, adjusting every layer may lead to overfitting due to the overwhelming number of parameters involved (Chollet, 2018). In our endeavor, the fine-tuning is carried out on different layers to evaluate and ascertain the most promising outcomes.

To determine the ideal CNN architecture for our project from the myriad of available configurations, we assessed four outstanding models suitable for a transfer learning approach. These are VGGNet (Simonyan and Zisserman, 2015), GoogLeNet (Szegedy et al., 2016), ResNet (He et al., 2016), and Xception (Chollet, 2017). All these CNN architectures have been trained on extensive image datasets and have achieved first place in the ImageNet image recognition competition, even exceeding human-level image recognition. Early stopping is utilized to mitigate potential overfitting. A smaller learning rate is deemed necessary to preserve previously learned features because transfer learning methods are used. In addition, the learning rate is reduced by 50% if no improvement is observed over five consecutive epochs. As a result, a relatively small learning rate of 0.001 is chosen to preserve pre-trained knowledge, ensure stable convergence, and reduce the risk of overfitting. To best serve our specific requirements, we further fine-tune the models deemed most appropriate. The process of tweaking hyperparameters is pivotal in refining model performance. Ultimately, we will analyze the effects of the hyperparameters.

To quantify the discrepancy between predicted and actual values of our CNN model, we adopted mean absolute error (MAE) as loss functions. MAE is the average of the absolute differences between predicted and actual values. Its magnitude reflects the degree to which the predicted values deviate from the actual ones. It has a smaller influence from outliers. Lower MAE indicates a better model prediction performance.

2.5 *Data processing for CNN models*

We first normalize the data, which is beneficial for enhancing the model's convergence rate and performance. The IR cloud images are consistently cropped down to a subimage of 256 × 256 pixels, with the side lengths of 1255 km × 1255 km centered on the eye of the TC. The total sample records of TC from 19 yr (2004–2022) are 4155. Training a deep learning model typically demands a substantial volume of data. To enlarge our initial dataset, we implement data augmentation techniques, specifically employing rotations of ±45 degrees to enrich the diversity of our image samples.

The data are further divided into training, validation, and test data. The training and validation data span from 2004 to 2019, while the test dataset is taken from 2020 to 2022. To enhance model accuracy, we employ the *k*-fold cross-validation method proposed by Geisser (1975) to divide the training and validation sets. We split the dataset into 5 equal parts, with each part used once as the validation set and the remaining parts used as the training set. This process is repeated 5 times, providing a more accurate estimation of the model performance. The sample numbers for the training, validation, and test datasets prior to augmentation are shown in Table 2.

3. Results

3.1 *Selection of the model architecture*

We first determine an optimal CNN architecture mentioned before as the base model (Simonyan and Zisserman, 2015; He et al., 2016; Szegedy et al., 2016; Chollet, 2017; Krizhevsky et al., 2017). Each model is adjusted and trained to minimize the MAE between the network's predictions and the TC intensity derived from JTWC. The outcomes are presented in Table 3.

It can be observed that the configuration yielding the best performance is the VGGNet model, and hence, it is chosen as the backbone for our application. This adjusted pre-trained VGGNet model starts with five stacks of convolutional layers, each utilizing a 3 × 3 receptive field, with max pooling layers strategically interspersed between these stacks to retain paramount meteorological features while simultaneously reducing computational demands. Following the last pooling layer, the model includes three fully connected (FC) layers. The original softmax output layer is replaced to suit regression tasks. The Rectified Linear Unit (ReLU) activation function is integrated within the model, accelerating convergence and enhancing its ability to decipher complex atmospheric

Table 2. Sample numbers of the training, validation, and test data. Due to the *k*-fold cross-validation method for splitting the training and validation datasets, they are considered as a whole

	Training and validation data	Test data
Sample number	3619	536

Table 3. Performance of different CNN frameworks. The results are derived from experiments on the test dataset. The model performance is gauged by MAE, with a lower MAE signifying superior performance. The minimum error is highlighted in bold typeface

CNN framework	MAE (m s^{-1})
VGGNet	**6.34**
GoogLeNet	10.34
ResNet	8.58
Xception	7.09

patterns. Batch normalization has been introduced on top of the pre-trained model to optimize the training process. Furthermore, a flatten layer is introduced, preserving all the intricate information in the feature maps while offering greater parameter capacity. Subsequently, the model employs multiple fully connected dense layers and dropout layers, specifically designed for regression-based predictions to reduce overfitting and improve the model's generalization ability on training data. Considering the dynamic nature of meteorological data, early stopping, L2 regularization, and dropout layers are implemented to avoid overfitting. By leveraging modern CNN strategies within a crafted architecture, our model is proficient in analyzing meteorological data, identifying vital atmospheric characteristics, and ensuring higher predictive accuracy in regression tasks. From Table 3, the VGGNet architecture demonstrates the highest performance in TC intensity estimations among the four models examined. However, it is worth noting that this model may not be universally applicable. Fine-tuning and hyperparameter adjustments may be imperative.

3.2 *Development of the model*

3.2.1 *Fine-tuned VGGNet architecture*

Based on the optimal CNN architecture, VGGNet, we utilize the fine-tuning method, which freezes the deeper convolutional layers and adapts the head of the network to the particular regression problem. The experimental results are presented in Table 4. The first column in Table 4 indicates the number of unfrozen layers in the VGGNet architecture. It can be illustrated that when only a few layers are unfrozen, the model performance on the test dataset for estimating TC intensity is inferior. As the number of unfrozen layers increases, the MAE generally decreases. Unfreezing and training the model's bottom layers allows the model to adjust those complex features that are closer to the specific task while keeping the generic features unchanged. Training only a limited number of layers might potentially constrain the model's adaptability, preventing it from effectively capturing and learning its features. But as the number of unfrozen layers increases, there is a slight rise in the MAE. There is a possibility that with too few layers frozen, most of the model parameters get updated, leading to overfitting despite employing early stopping and dropout layers. This is especially possible when the training dataset is small, resulting in poorer prediction performance on test data. Unfreezing and training only a few layers can reduce this risk, allowing the model to be finely tuned for the specific task.

Table 4. Performance of the fine-tuned VGGNet architecture with different bottom layers left unfrozen. The minimum error is highlighted in bold typeface

Unfrozen layer	MAE (m s^{-1})
1	6.45
2	6.52
3	6.46
4	6.48
5	6.12
7	**5.98**
10	6.43
15	6.58

In conclusion, through numerous experiments unfreezing different layers, a relatively suitable fine-tuning scheme has been identified, allowing the model to both leverage the pre-trained weights and adapt to our task. Note that when unfreezing 7 layers, our model more effectively extracts and reorders potential features of TC intensity, obtaining an MAE as low as 5.98 m s^{-1}, which is considered to be a satisfactory and comparable result.

3.2.2 *Temporal enhancement by sequential images*

Based on the above results, it is evident that the VGGNet architecture performs well in capturing the complex relationship between the IR cloud imagery and TC intensity data. However, the current algorithm is mostly similar to image recognition methods used in computer science and does not include specific weather-related knowledge. A series of IR cloud images taken over time can provide crucial dynamic information on TC evolution, such as the formation, growth, relocation, and decay of convective cloud clusters, which are intricately linked to TC intensity. Such dynamic information is not discernible from a standalone IR image, making time-sequential data more illuminating for understanding TC intensity characteristics. Thus, temporal enhancement is introduced into our VGGNet model to provide additional contextual information and enhance its ability to capture the complex spatial and temporal distributions of convection, which is beneficial for indicating TC intensity. Moreover, employing multiple time-sequential images mitigates the risk of missing satellite observations at any specific time. Given computational constraints and the model's potential utility for practical applications, coupled with the aforementioned benefits, we opt to use a sequence of satellite images spanning 12 h prior to the current time at 6-h intervals. The results are detailed in Table 5.

As evidenced by Table 5, when utilizing a continuous time series from −12 h to the current, our model successfully captures the feature of TC intensity with an MAE of 5.48 m s^{-1}. The test set error shows an 8% improvement relative to the experiments without sequential images, thereby proving the effectiveness of using temporal

Table 5. Performance of the VGGNet architecture with successive IR image sequences of different time spans. The minimum error is highlighted in bold typeface

Time span	MAE (m s^{-1})
−6 h to current	5.77
−12 h to current	**5.48**

factors for elevating the model's performance. Owing to recent advancements in satellite technology, the *Himawari-8* satellite is now equipped to provide data with enhanced spatiotemporal resolution. This technological progress paves the way for the future integration of higher-resolution continuous cloud imagery, offering the potential to further improve the performance of the model.

3.2.3 *Incorporating physical information*

Based on the aforementioned experimental results, we have discerned that after a series of optimization processes, given a set of satellite IR cloud images, we can employ the VGGNet model to estimate TC intensity with considerable accuracy. Currently, in the field of TC research, a significant amount of physical information regarding TC intensity has been explicitly collected and validated. However, this has not been fully integrated into deep learning models. Given that the physical knowledge of TCs could play a critical role in deep learning algorithms for intensity estimation, it can furnish the model with information that may be non-existent or challenging to extract from the restricted training data. This not only facilitates improved training efficiency and enhanced model generalization but also fortifies the physical plausibility of the model. While the incorporation of sequential images has already integrated certain weather-related knowledge, to elevate the accuracy in estimating TC intensity, there is a prevailing necessity for the current model to assimilate additional physical information. Such an integration will be the key point of our subsequent research.

The evolution of TC intensity is a complex process, primarily driven by three key factors: environmental conditions, internal structure and dynamics of TC, and interaction between TC and ocean (Marks et al., 1998; Emanuel, 1999). Among these factors, specific parameters like VWS, HR, and OHC stand out not only for their strong association with the intensity of the TC but also for their capability to enhance or supplement the features that the VGGNet model extracts from satellite cloud images (Kaplan and DeMaria, 2003; Kaplan et al., 2010, 2015; Shu et al., 2012). The intensification of a TC predominantly hinges on the energy that it derives from warm sea waters. Thus, a higher OHC tends to favor TC strengthening (Riehl, 1950). Nevertheless, the interplay between TC and ocean is additionally influenced by the translation speed of the TC. As the speed decreases, the interaction begins to become more important, having a significant impact on the intensity of the TC (Price, 1981; Lloyd and Vecchi, 2011). Besides, as pointed out by Zhao and Chan (2017), a TC's potential for strengthening escalates with its speed, since rapid movement can counterbalance the negative effects of ocean cooling. This suggests that there might be a subtle yet crucial interaction between OHC and the TC translation speed. It is not merely the individual effects of OHC and TC speed that matter; their combined interplay also plays a significant role in influencing TC intensity. Additionally, the TC duration is also a key factor.

Considering the aforementioned background and insights, we further explore the effect of integrating more related physical information and their different combinations into the VGGNet model. The following parameters are particularly focused on: VWS, RH, OHC, TC translation speed, and duration. The main results are shown in Table 6.

From the results, it can be observed that upon the addition of physical information, all parameters, except for RH, reduce the MAE value. This may be attributed to the fact that the TCs in this study primarily occur over the open ocean, where RH is generally high. Therefore, introducing this particular parameter alone does not markedly enhance the model's predictive accuracy. However, the inclusion of other physical information aids in bolstering the model performance, albeit to varying extents. To specify, VWS shows a modest improvement to the model, with only about a 4% enhancement compared to the VGGNet model without any additional physical information (control model). This may be attributed to the inherent sophistication of the VGGNet model, which is adept at discerning intricate features correlated with VWS from satellite IR imagery, particularly from the distribution of convective cloud clusters. In terms of

Table 6. Performance of the VGGNet architecture with different combinations of the physical information. The minimum error is highlighted in bold typeface

Physical information	MAE (m s^{-1})
RH	5.56
OHC	5.40
Translation speed	5.28
VWS	5.26
OHC + translation speed + RH	5.14
OHC + translation speed	5.10
Duration	5.03
All combinations without VWS	4.87
All combinations	4.80
OHC + translation speed + VWS	4.77
All combinations without RH	4.73
OHC + translation speed + duration	**4.65**

OHC, although it is widely believed that elevated OHC typically promotes TC intensification, in areas prone to frequent TC formation, high OHC is inherently present. However, the actual intensities of TCs tend to be lower in these regions. Furthermore, given the interplay between OHC and translation speed, relying solely on OHC as a physical parameter might not be sufficient. As can be seen from the outcomes, when both OHC and translation speed are incorporated, the model's MAE drops to 5.10 m s^{-1}, reflecting a 7% improvement over the control model and surpassing the results that only OHC or translation speed is considered.

In comparison, the model shows a more pronounced performance when only incorporating the TC duration. This may be attributed to the fact that a TC's duration potentially reflects the general characteristics of TC intensity evolution. The model, influenced by the initial phases of storm age, is inclined to offer more conservative, lower-intensity estimates rather than commencing with elevated values. The model achieves its optimal performance with the MAE of 4.65 m s^{-1}, when it incorporates OHC, TC translation speed, and duration. These features directly link to the energy sources and structural changes of TCs, establishing a physical correlation between the satellite IR images and TC intensity output. While the inclusion of VWS moderately improves model performance, integrating VWS with OHC, TC translation speed, and duration reduces overall effectiveness. This decrease may stem from the complexity introduced by VWS, which could exceed the learning capacity of the current dataset, thereby not achieving the anticipated performance improvements. These insights pinpoint prospective research trajectories, focusing on the effective incorporation of such information to further optimize model performance.

3.2.4 *Adjusting the hyperparameters*

The settings for hyperparameters, such as learning rate and batch size, can vary widely depending on the specific tasks at hand. Given the complexity and spatiotemporal variability of the satellite IR image data, it becomes imperative to adjust these parameters, which will enable the VGGNet model to more effectively learn the features of TC intensity across various scales, thereby enhancing the accuracy of its predictions. From the previous sections, two experiments with relatively low MAE are selected: one using OHC, TC translation speed combined with duration, and the other involving all combinations except for RH. For each of these setups, we adjust the learning rates and batch sizes to determine the optimal hyperparameters that yield the best model performance. The outcomes of the model after these adjustments are presented in Table 7.

As depicted in Table 7, the empirical results clearly highlight that the model tends to yield a lower MAE when the learning rate is adjusted to 0.0001, paired with a batch size of either 16 or 32. Broadly speaking, integrating OHC, TC translation speed, and duration as the auxiliary physical information enhances the model performance. When fine-tuned with these specific hyperparameters, the model achieves its peak performance for TC intensity estimation, yielding a minimum MAE of 4.05 m s^{-1}. The selection of these hyperparameters likely facilitates a balance between swift convergence and maintaining model stability. Based on these empirical outcomes, we have opted for this particular set of hyperparameters as our final model configuration. While acknowledging the extensive range of possible hyperparameter combinations, we affirm that although our model may not represent the optimal architecture, it nonetheless exhibits strong performance in practical applications, thereby confirming its utility.

3.3 *Performance evaluation*

The optimal VGGNet model is crafted by combining a range of beneficial training techniques. The settings incorporate the rotation augmentation of the initial data, and the application of transfer learning in conjunction with fine-tuning methodologies. Moreover, the use of auxiliary physical information such as VWS, OHC, RH, TC duration, and translation speed provides valuable insights. Additionally, temporal enhancement is achieved by using IR sequence images from the past 12 h, 6 h, and the current period. The adjustment of hyperparameters further enhances the model performance.

Figure 1 denotes the promising outcomes by comparing the estimated intensity derived from our final VGGNet model with the TC intensity as recorded in the

Table 7. Performance of the VGGNet architectures with different learning rates (LR) and batch sizes (BS) using (a) OHC, TC translation speed combined with duration, and (b) all combinations without RH as the physical knowledge. The minimum error is highlighted in bold typeface

BS	LR			
	0.0001	0.0005	0.001	0.005
(a) OHC, TC translation speed combined with duration MAE (m s^{-1})				
8	4.05	4.35	4.45	5.28
16	4.12	4.34	4.42	5.53
32	**4.05**	4.40	4.65	5.25
64	4.46	4.79	5.42	6.05
(b) All combinations without RH MAE (m s^{-1})				
8	4.25	4.44	4.46	5.26
16	**4.16**	4.47	4.41	5.12
32	4.19	4.41	4.73	5.11
64	4.59	5.12	5.25	6.54

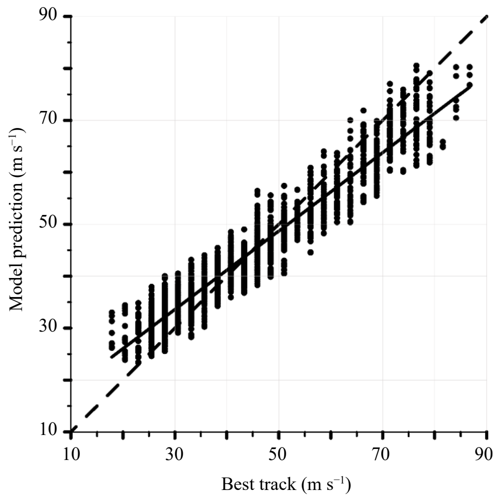

Fig. 1. Scatterplot comparing the TC intensity (m s^{-1}) from the VGGNet model with the best track records for TCs over the western North Pacific during 2020–2022. The blue line represents the linear fit, while the black line indicates where the VGGNet model prediction matches the best track data.

best track data. An impressive correlation is evident from the linear fitting, showcasing that the intensities projected by the VGGNet model are in high concordance with the best track data. However, there are also some deviations. When the intensity drops below 28.05 m s^{-1}, there is an overestimation trend. On the contrary, TCs with intensities exceeding 58.65 m s^{-1} exhibit a propensity to be underestimated by our model. This may arise from the scarcity of intense TC cases used for training. Nevertheless, considering that a substantial proportion of TCs falls between 30.6 to 45.9 m s^{-1}, the variations observed for these outlier intensities are deemed acceptable.

3.4 *Comparison with other techniques*

The comparative analysis with existing TC intensity estimation models, as detailed in Table 8, demonstrates the superior performance of our model. Our results surpass those derived from traditional machine learning techniques, such as Estimation TC Intensity (ETCI; Liu et al., 2019), PIEAS HURricane Intensity Estimator-Support Vector Regression (PHURIE-SVR; Asif et al., 2020), and Relevance Vector Machine + Deviation Angle Gradient Co-Occurrence Matrix (RVM + DAGCOM; Zhang et al., 2019). Additionally, our model also outperforms some contemporary CNN-based models like Deep-PHURIE model (Dawood et al., 2020) that uses infrared satellite imagery to reduce manual labeling. Unlike the Multiple Linear Regression-Brightness Temperature-Sea Surface Wind speed (MLR-TB-SSW) model by Yang and Xiang (2019), which integrates sea surface wind speed and Special Sensor Microwave Imager/Sounder (SSMIS) microwave radiometer data, our model relies solely on imagery from a single satellite platform. Furthermore, our model slightly outperforms the well-established enhanced form of the Dvorak technique, i.e., ADT9.0, which incorporates aircraft observations, microwave data, and multispectral satellite imagery for operational use as improved by Olander and Velden (2019). In conclusion, our model presents lower MAE values in intensity estimation compared to traditional machine learning approaches, certain CNN models, operational techniques, and models utilizing diverse datasets, highlighting its potential as an effective and promising TC intensity estimation method.

Table 8. Comparison of the best performance in this paper with other referenced models

Model	MAE (m s^{-1})	Reference
PHURIE-SVR	7.21	Asif et al. (2020)
Deep-PHURIE	4.27	Dawood et al. (2020)
ADT9.0	4.42	Olander and Velden (2019)
MLR-TB-SSW	4.62	Yang and Xiang (2019)
ETCI	7.16	Liu et al. (2019)
RVM + DAGCOM	7.37	Zhang et al. (2019)
Multiple linear regression model	4.85	Zhao et al. (2016)

3.5 *Interoperability of the model*

Traditional TC intensity forecasting has primarily been based on human understanding of TCs and manual analysis of specific features, such as spiral patterns, TC core regions, and asymmetry. However, conventional techniques overlook potential features that have yet to be identified but could be more relevant to TCs. Fundamentally, these regular methods remain subjective. CNN can extract a broader range of features from TC images and establish a relationship between these features and TC intensity. Although there are no tools for intuitive analysis of the results derived from CNNs, the CNN approach is objective. By combining information extracted from CNNs, we may uncover some previously unknown attributes that impact TCs.

However, similar to most deep learning methods, our final VGGNet model, whose core algorithm is based on CNNs, is often perceived as a "black box." Understanding why the model produces a particular prediction and its credibility is essential to uncover the potential shortcomings of the model. Besides, such insights can enhance our understanding of the model and aid in its further refinement. To address this interpretability challenge, we employ saliency maps. The SmoothGrad combined with the Integrated Gradients approach (Smilkov et al., 2017) is utilized. This method can reflect which features of the input are responsible for the model's output. The combination of SmoothGrad and Integrated Gradients takes advantage of the noise-reduction capabilities of the former and the path-based attribution of the latter, yielding saliency maps that are clearer, more consistent, and more interpretable than using either method alone. The saliency map illustrates the influence of each pixel in the image on the final intensity estimation, based on their relative gradients to the output. Its value can be positive or negative, indicating a positive or negative influence. The larger the absolute value in the map, the more significant the contribution of the corresponding pixel in the cloud image to the final VGGNet model's TC intensity estimate.

Figure 2a showcases the formation phase of TC Lionrock. Figure 2b marks the point where Lionrock nears the end of its intensification, during which it maintains a robust speed of around 43.35 m s^{-1} for an extended duration. By the time in Fig. 2c, the TC begins its weakening phase, and in Fig. 2d, Lionrock is significantly weakened and on the brink of dissipating. Notably, the intensity predicted by the VGGNet model closely matches the data from the best track. It can be illustrated that the saliency maps reveal the comprehensive structure of a TC, highlighting a prominent inner core at the image's center. This model skillfully captures the cloud system distributions from the saliency values shown in Fig. 2. It underscores that the TC intensity estimation is primarily driven by the TC inner core and further shaped by the combination of outer spiral rainbands and convective systems.

The interpretation of the saliency maps is grounded in meteorological principles. In cloud imagery, the arrangement of cloud systems can reflect the energy and moisture influx from the ocean, and it can also potentially provide insights into VWS. Moreover, the decentration between convections and the TC center can indicate VWS. Thus, the configuration and strength of the cloud systems serve as pivotal indicators of modulating TC intensity. Disorganized, fragmented, or dispersed cloud patterns often hint at a weakening TC, while cohesive and well-structured systems suggest strengthening conditions. The saliency maps further enhance our understanding. The results suggest that while our VGGNet model's prediction of TC intensity from IR cloud images is data-driven, the deep learning model indeed captures physically meaningful configurations of the TC cloud systems. Thus, employing deep learning methods might be beneficial for probing more intricate TC mechanisms, and further research along with more advanced and appropriate interpretation techniques is required.

4. Conclusions

TCs are one of the most severe disasters. Estimating their intensity is crucial for forecasters and institutions to issue disaster warnings. This paper develops a unique CNN architecture to extract TC intensity values from the geostationary satellite IR imagery over the western North Pacific Ocean basin during 2004–2022. The methodology offers a dependable foundation for TC intensity prediction systems. We incorporate the pre-trained weights

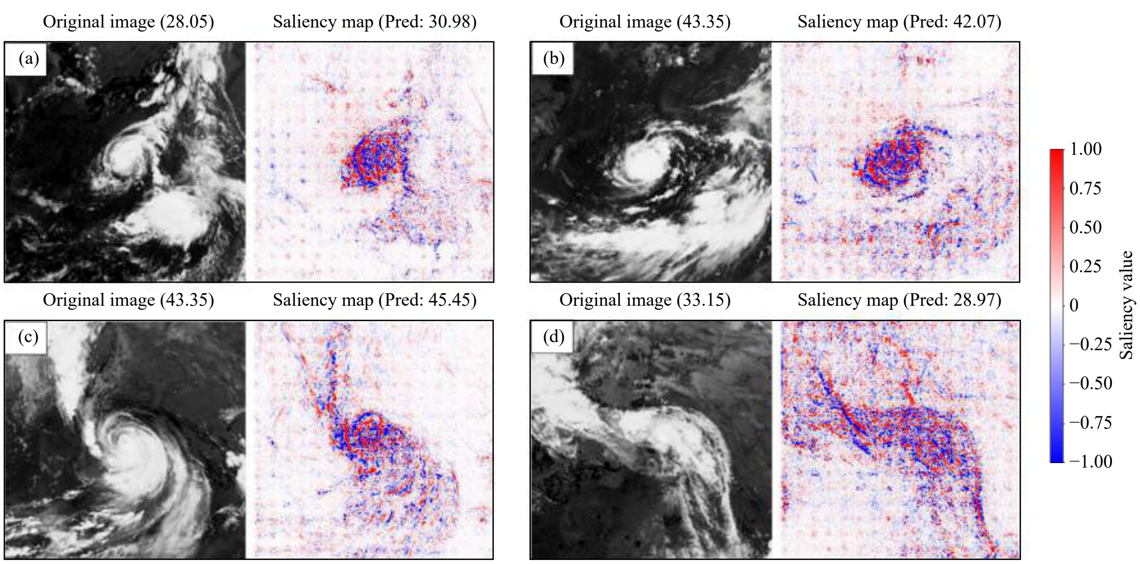

Fig. 2. IR satellite cloud images (the first and third columns) and their corresponding saliency maps (the second and fourth columns) for TC Lionrock (1610) at (a) 0600 UTC 23 August 2016, (b) 0000 UTC 25 August 2016, (c) 0000 UTC 29 August 2016, and (d) 0600 UTC 30 August 2016. The TC intensity (m s^{-1}) indicated in the parentheses following the original images is from the best track, while the predicted TC intensity (m s^{-1}) is displayed subsequent to the saliency maps.

of four prominent CNN architectures and subsequently fine-tune them to meet our specific task. Techniques such as the rotation augmentation of the initial datasets, temporal enhancement via sequential imagery, incorporating auxiliary physical information, and hyperparameter tuning are employed, resulting in an optimized VGGNet model. The findings demonstrate a superior representation of TC intensity, as inferred from IR imagery, obtaining a minimum MAE of 4.05 m s^{-1}.

To further enhance the interpretability of the model, we combine the SmoothGrad and Integrated Gradients methods to obtain a clearer, more coherent, and easily interpretable saliency map. These maps reveal the input features considered by the model when predicting TC intensity. The saliency maps further unveil that the overall structure of the TC, especially its distinct inner core region, plays a pivotal role in intensity prediction. The configuration of the cloud system also provides the model with key insights into the current state and potential development of the TC. In summary, the saliency maps offer us a deep understanding of how the VGGNet model assesses TC intensity, showcasing the contributions of different cloud system features to the prediction.

The marked enhancement in prediction underscores the efficacy of our chosen VGGNet model. The numerous enhancements implemented in our model's performance provide valuable perspectives for forecasting TC intensity and by extension, other TC metrics. This bears significant implications for future practical applications, potentially mitigating the detrimental impacts of increasingly severe TC disasters. However, our models are trained and evaluated by using data available up to 2022. The performance of the model with the addition of new data remains unpredictable. To ensure sustained accuracy, it is advisable to update the parameters and weights of the various algorithms before deploying the model in practical applications. While our outcomes are promising, it is important to acknowledge the intricate factors governing convective intensities and distributions. Besides, considering that our VGGNet model estimates TC intensity using only IR satellite cloud image data, further enhancing accuracy with multi-source data might also be beneficial.

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Appendix

Leipper and Volgenau (1972) introduced a new concept, termed OHC. They point out that using SST to describe the latent heat flux is not accurate. OHC is a more accurate and superior factor. The formula for calculating OHC is:

$$\mathrm{OHC} = c_p \int_0^{D_{26}} \rho\,[T(z) - 26]\,\mathrm{d}z, \tag{A1}$$

where c_p is the specific heat at constant pressure, and ρ is the seawater density. Its value is usually set as a constant, but it is also related to the salinity within the ocean. D_{26} represents the thickness of the 26°C isotherm layer, and T is the seawater temperature, which changes with the water depth (Shay and Brewster, 2010). Theoretically, the larger the OHC value, the thicker the warm water layer above 26°C, which is more conducive to the formation or intensification of TCs.

REFERENCES

Asif, A., M. Dawood, B. Jan, et al., 2020: PHURIE: Hurricane intensity estimation from infrared satellite imagery using machine learning. *Neural Comput. Appl.*, **32**, 4821–4834, doi: 10.1007/s00521-018-3874-6.

Bessho, K., K. Date, M. Hayashi, et al., 2016: An introduction to Himawari-8/9—Japan's new-generation geostationary meteorological satellites. *J. Meteor. Soc. Japan*, **94**, 151–183, doi: 10.2151/jmsj.2016-009.

Bolton, T., and L. Zanna, 2019: Applications of deep learning to ocean data inference and subgrid parameterization. *J. Adv. Model. Earth Syst.*, **11**, 376–399, doi: 10.1029/2018MS001472.

Chen, B., B.-F. Chen, and H.-T. Lin, 2018: Rotation-blended CNNs on a new open dataset for tropical cyclone image-to-intensity regression. Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, ACM, London, UK, 90–99, doi: 10.1145/3219819.3219926.

Chen, R., W. M. Zhang, and X. Wang, 2020: Machine learning in tropical cyclone forecast modeling: A review. *Atmosphere*, **11**, 676, doi: 10.3390/atmos11070676.

Chollet, F., 2017: Xception: Deep learning with depthwise separable convolutions. Proceedings of 2017 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Honolulu, HI, USA, 1251–1258, doi: 10.1109/CVPR.2017.195.

Chollet, F., 2018: *Deep Learning with Python*. Manning Publications Co., Shelter Island, 306 pp.

Dawood, M., A. Asif, and F. U. A. A. Minhas, 2020: Deep-PHURIE: Deep learning based hurricane intensity estimation from infrared satellite imagery. *Neural Comput. Appl.*, **32**, 9009–9017, doi: 10.1007/s00521-019-04410-7.

Emanuel, K. A., 1999: Thermodynamic control of hurricane intensity. *Nature*, **401**, 665–669, doi: 10.1038/44326.

Fetanat, G., A. Homaifar, and K. R. Knapp, 2013: Objective tropical cyclone intensity estimation using analogs of spatial features in satellite data. *Wea. Forecasting*, **28**, 1446–1459, doi: 10.1175/WAF-D-13-00006.1.

Geisser, S., 1975: The predictive sample reuse method with applications. *J. Amer. Stat. Assoc.*, **70**, 320–328, doi: 10.1080/01621459.1975.10479865.

Ham, Y. G., J. H. Kim, and J. J. Luo, 2019: Deep learning for multi-year ENSO forecasts. *Nature*, **573**, 568–572, doi: 10.1038/s41586-019-1559-7.

Han, Y. L., G. J. Zhang, X. M. Huang, et al., 2020: A moist physics parameterization based on deep learning. *J. Adv. Model. Earth Syst.*, **12**, e2020MS002076, doi: 10.1029/2020MS002076.

He, K. M., X. Y. Zhang, S. Q. Ren, et al., 2016: Deep residual learning for image recognition. Proceedings of 2016 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Las Vegas, NV, USA, 770–778, doi: 10.1109/CVPR.2016.90.

Hubel, D. H., and T. N. Wiesel, 1962: Receptive fields, binocular interaction and functional architecture in the cat's visual cortex. *J. Physiol.*, **160**, 106–154, doi: 10.1113/jphysiol.1962.sp006837.

Jin, S. H., S. Wang, and X. F. Li, 2014: Typhoon eye extraction with an automatic SAR image segmentation method. *Int. J. Remote Sens.*, **35**, 3978–3993, doi: 10.1080/01431161.2014.916447.

Jin, S. H., S. Wang, X. F. Li, et al., 2017: A salient region detection and pattern matching-based algorithm for center detection of a partially covered tropical cyclone in a SAR image. *IEEE Trans. Geosci. Remote Sens.*, **55**, 280–291, doi: 10.1109/TGRS.2016.2605766.

Jin, S. H., X. F. Li, X. F. Yang, et al., 2019: Identification of tropical cyclone centers in SAR imagery based on template matching and particle swarm optimization algorithms. *IEEE Trans. Geosci. Remote Sens.*, **57**, 598–608, doi: 10.1109/TGRS.2018.2863259.

Judt, F., and S. S. Chen, 2016: Predictability and dynamics of tropical cyclone rapid intensification deduced from high-resolution stochastic ensembles. *Mon. Wea. Rev.*, **144**, 4395–4420, doi: 10.1175/MWR-D-15-0413.1.

Kaplan, J., and M. DeMaria, 2003: Large-scale characteristics of rapidly intensifying tropical cyclones in the North Atlantic basin. *Wea. Forecasting*, **18**, 1093–1108, doi: 10.1175/1520-0434(2003)018<1093:LCORIT>2.0.CO;2.

Kaplan, J., M. DeMaria, and J. A. Knaff, 2010: A revised tropical cyclone rapid intensification index for the Atlantic and eastern North Pacific basins. *Wea. Forecasting*, **25**, 220–241, doi: 10.1175/2009WAF2222280.1.

Kaplan, J., C. M. Rozoff, M. DeMaria, et al., 2015: Evaluating environmental impacts on tropical cyclone rapid intensification predictability utilizing statistical models. *Wea. Forecasting*, **30**, 1374–1396, doi: 10.1175/WAF-D-15-0032.1.

Kossin, J. P., J. A. Knaff, H. I. Berger, et al., 2007: Estimating hurricane wind structure in the absence of aircraft reconnaissance. *Wea. Forecasting*, **22**, 89–101, doi: 10.1175/WAF985.1.

Krizhevsky, A., I. Sutskever, and G. E. Hinton, 2017: ImageNet classification with deep convolutional neural networks. *Commun. ACM*, **60**, 84–90, doi: 10.1145/3065386.

Lagerquist, R., A. McGovern, and D. J. Gagne II, 2019: Deep learning for spatially explicit prediction of synoptic-scale fronts. *Wea. Forecasting*, **34**, 1137–1160, doi: 10.1175/WAF-D-18-0183.1.

LeCun, Y., Y. Bengio, and G. Hinton, 2015: Deep learning. *Nature*, **521**, 436–444, doi: 10.1038/nature14539.

Leipper, D. F., and D. Volgenau, 1972: Hurricane heat potential of the Gulf of Mexico. *J. Phys. Oceanogr.*, **2**, 218–224, doi: 10.1175/1520-0485(1972)002<0218:HHPOTG>2.0.CO;2.

Li, X. F., B. Liu, G. Zheng, et al., 2020: Deep-learning-based information mining from ocean remote-sensing imagery. *Natl. Sci. Rev.*, **7**, 1584–1605, doi: 10.1093/nsr/nwaa047.

Liu, J., X. F. Xu, and X. Y. Luo, 2019: Estimation of tropical cyclone intensity using infrared data from a geostationary satellite. *SOLA*, **15**, 189–192, doi: 10.2151/sola.2019-034.

Liu, Y. J., E. Racah, Prabhat, et al., 2016: Application of deep convolutional neural networks for detecting extreme weather in climate datasets. Proceedings of 2016 International Conference on Advances in Big Data Analytics, Computer Vision and Pattern Recognition, Boston, MA, USA, 81–88.

Lloyd, I. D., and G. A. Vecchi, 2011: Observational evidence for oceanic controls on hurricane intensity. *J. Climate*, **24**, 1138–1153, doi: 10.1175/2010JCLI3763.1.

Marks, F. D., L. K. Shay, G. Barnes, et al., 1998: Landfalling tropical cyclones: Forecast problems and associated research opportunities. *Bull. Amer. Meteor. Soc.*, **79**, 305–323, doi: 10.1175/1520-0477(1998)079<0305:LTCFPA>2.0.CO;2.

McGovern, A., R. Lagerquist, D. J. Gagne II, et al., 2019: Making the black box more transparent: Understanding the physical implications of machine learning. *Bull. Amer. Meteor. Soc.*, **100**, 2175–2199, doi: 10.1175/BAMS-D-18-0195.1.

Olander, T. L., and C. S. Velden, 2007: The advanced Dvorak technique: Continued development of an objective scheme to estimate tropical cyclone intensity using geostationary infrared satellite imagery. *Wea. Forecasting*, **22**, 287–298, doi: 10.1175/WAF975.1.

Olander, T. L., and C. S. Velden, 2019: The advanced Dvorak technique (ADT) for estimating tropical cyclone intensity: Update and new capabilities. *Wea. Forecasting*, **34**, 905–922, doi: 10.1175/WAF-D-19-0007.1.

Pan, S. J., and Q. Yang, 2010: A survey on transfer learning. *IEEE Trans. Knowl. Data Eng.*, **22**, 1345–1359, doi: 10.1109/TKDE.2009.191.

Piñeros, M. F., E. A. Ritchie, and J. S. Tyo, 2011: Estimating tropical cyclone intensity from infrared image data. *Wea. Forecasting*, **26**, 690–698, doi: 10.1175/WAF-D-10-05062.1.

Pradhan, R., R. S. Aygun, M. Maskey, et al., 2018: Tropical cyclone intensity estimation using a deep convolutional neural network. *IEEE Trans. Image Process.*, **27**, 692–702, doi: 10.1109/TIP.2017.2766358.

Price, J. F., 1981: Upper ocean response to a hurricane. *J. Phys. Oceanogr.*, **11**, 153–175, doi: 10.1175/1520-0485(1981)011<0153:UORTAH>2.0.CO;2.

Ramdan, A., A. Heryana, A. Arisal, et al., 2020: Transfer learning and fine-tuning for deep learning-based tea diseases detection on small datasets. Proceedings of 2020 International Conference on Radar, Antenna, Microwave, Electronics, and Telecommunications, IEEE, Tangerang, Indonesia, 206–211, doi: 10.1109/ICRAMET51080.2020.9298575.

Riehl, H., 1950: A model of hurricane formation. *J. Appl. Phys.*, **21**, 917–925, doi: 10.1063/1.1699784.

Ritchie, E. A., G. Valliere-Kelley, M. F. Piñeros, et al., 2012: Tropical cyclone intensity estimation in the North Atlantic basin using an improved deviation angle variance technique. *Wea. Forecasting*, **27**, 1264–1277, doi: 10.1175/WAF-D-11-00156.1.

Shay, L. K., and J. K. Brewster, 2010: Oceanic heat content variability in the eastern Pacific Ocean for hurricane intensity forecasting. *Mon. Wea. Rev.*, **138**, 2110–2131, doi: 10.1175/2010MWR3189.1.

Shu, S. J., J. Ming, and P. Chi, 2012: Large-scale characteristics and probability of rapidly intensifying tropical cyclones in the western North Pacific basin. *Wea. Forecasting*, **27**, 411–423, doi: 10.1175/WAF-D-11-00042.1.

Simonyan, K., and A. Zisserman, 2015: Very deep convolutional networks for large-scale image recognition. Proceedings of the 3rd International Conference on Learning Representations, IEEE, San Diego, USA, 313–318.

Smilkov, D., N. Thorat, B. Kim, et al., 2017: SmoothGrad: Removing noise by adding noise. Available online at doi: 10.48550/arXiv.1706.03825. Accessed on 5 March 2024.

Sønderby, C. K., L. Espeholt, J. Heek, et al., 2020: MetNet: A neural weather model for precipitation forecasting. Available online at doi: 10.48550/arXiv.2003.12140. Accessed on 5 March 2024.

Szegedy, C., V. Vanhoucke, S. Ioffe, et al., 2016: Rethinking the Inception architecture for computer vision. Proceedings of 2016 IEEE Conference on Computer Vision and Pattern Recognition, IEEE, Las Vegas, NV, USA, 2818–2826, doi: 10.1109/CVPR.2016.308.

Takeuchi, Y., 2018: An introduction of advanced technology for tropical cyclone observation, analysis and forecast in JMA. *Trop. Cyclone Res. Rev.*, **7**, 153–163, doi: 10.6057/2018TCRR03.01.

Velden, C., B. Harper, F. Wells, et al., 2006: The Dvorak tropical cyclone intensity estimation technique: A satellite-based method that has endured for over 30 years. *Bull. Amer. Meteor. Soc.*, **87**, 1195–1210, doi: 10.1175/BAMS-87-9-1195.

Velden, C., T. Olander, D. Herndon, et al., 2017: Reprocessing the most intense historical tropical cyclones in the satellite era using the advanced Dvorak technique. *Mon. Wea. Rev.*, **145**, 971–983, doi: 10.1175/MWR-D-16-0312.1.

Wang, C., G. Zheng, X. F. Li, et al., 2022: Tropical cyclone intensity estimation from geostationary satellite imagery using deep convolutional neural networks. *IEEE Trans. Geosci. Remote Sens.*, **60**, 1–16, doi: 10.1109/TGRS.2021.3066299.

Webster, P. J., G. J. Holland, J. A. Curry, et al., 2005: Changes in tropical cyclone number, duration, and intensity in a warming environment. *Science*, **309**, 1844–1846, doi: 10.1126/science.1116448.

Weyn, J. A., D. R. Durran, and R. Caruana, 2020: Improving data-driven global weather prediction using deep convolutional neural networks on a cubed sphere. *J. Adv. Model. Earth Syst.*, **12**, e2020MS002109, doi: 10.1029/2020MS002109.

Wu, L. G., and H. K. Zhao, 2012: Dynamically derived tropical cyclone intensity changes over the western North Pacific. *J. Climate*, **25**, 89–98, doi: 10.1175/2011JCLI4139.1.

Xu, Q., G. S. Zhang, X. F. Li, et al., 2016: An automatic method for tropical cyclone center determination from SAR. Proceedings of 2016 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), IEEE, Beijing, China, 2250–2252, doi: 10.1109/IGARSS.2016.7729581.

Yang, X. F., and K. S. Xiang, 2019: Synergistic use of satellite active and passive microwave observations to estimate typhoon intensity. Proceedings of 2019 Photonics & Electromagnetics Research Symposium-Spring (PIERS-Spring), IEEE, Rome, Italy, 1612–1617, doi: 10.1109/PIERS-Spring46901.2019.9017799.

Zahera, H. M., M. A. Sherif, and A. C. N. Ngomo, 2019: Jointly learning from social media and environmental data for typhoon intensity prediction. Proceedings of the 10th International Conference on Knowledge Capture, ACM, Marina Del Rey, CA, USA, 231–234, doi: 10.1145/3360901.3364413.

Zhang, C.-J., Q. Luo, L.-J. Dai, et al., 2019: Intensity estimation of tropical cyclones using the relevance vector machine from infrared satellite image data. *IEEE J. Sel. Top. Appl. Earth Obs. Remote Sens.*, **12**, 763–773, doi: 10.1109/JSTARS.2019.2894654.

Zhao, X. H., and J. C. L. Chan, 2017: Changes in tropical cyclone intensity with translation speed and mixed-layer depth: Idealized WRF-ROMS coupled model simulations. *Quart. J. Roy. Meteor. Soc.*, **143**, 152–163, doi: 10.1002/qj.2905.

Zhao, Y., C. F. Zhao, R. Y. Sun, et al., 2016: A multiple linear regression model for tropical cyclone intensity estimation from satellite infrared images. *Atmosphere*, **7**, 40, doi: 10.3390/atmos7030040.

Zheng, G., J. S. Yang, A. K. Liu, et al., 2016: Comparison of typhoon centers from SAR and IR images and those from best track data sets. *IEEE Trans. Geosci. Remote Sens.*, **54**, 1000–1012, doi: 10.1109/TGRS.2015.2472282.

Zhuge, X.-Y., J. Guan, F. Yu, et al., 2015: A new satellite-based indicator for estimation of the western North Pacific tropical cyclone current intensity. *IEEE Trans. Geosci. Remote Sens.*, **53**, 5661–5676, doi: 10.1109/TGRS.2015.2427035.

Zhuo, J.-Y., and Z.-M. Tan, 2021: Physics-augmented deep learning to improve tropical cyclone intensity and size estimation from satellite imagery. *Mon. Wea. Rev.*, **149**, 2097–2113, doi: 10.1175/MWR-D-20-0333.1.

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